

School Legacy: Reconstructing Memory Through AI-augmented Social XR

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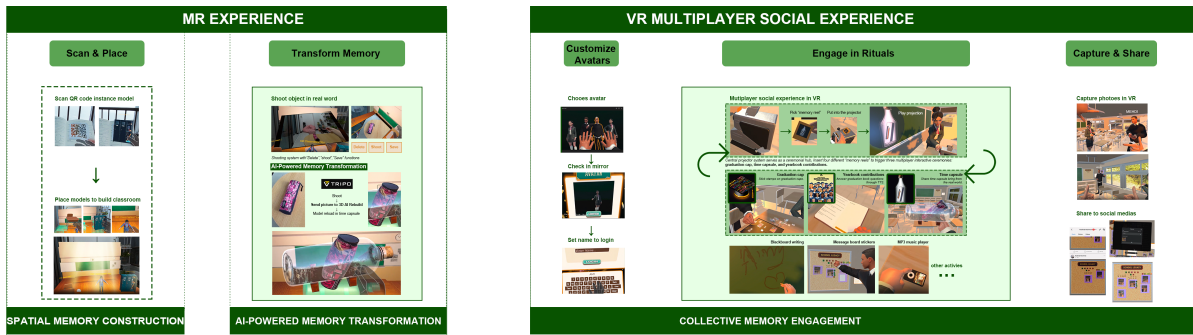


Figure 1: System Design

Abstract

School Legacy is an interactive Extended Reality (XR) experience that transforms personal school memories into a shared, AI-augmented virtual classroom. By scanning a physical memory book, users unlock digital artifacts and generate 3D "time capsules" from personal photos using generative AI. These capsules serve as emotional keys to enter a multiplayer VR environment where alumni reconnect through voice, gesture, and nostalgic rituals—such as decorating graduation caps and signing digital yearbooks. The system blends tangible interaction, real-time co-presence, and AI-driven content creation to create a living archive of youth, turning private nostalgia into a collective, embodied experience.

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1 Introduction

The classroom often remains frozen in memory, a spatial container of shared youth that becomes increasingly unreachable with time. While photographs preserve isolated moments, they fail to capture the lived spatial and social dimensions of school life, reducing communal experiences to fragmented records. For international students who left their home countries at a young age, this challenge is even greater: formative memories remain tied to distant places, and geographically dispersed classmates make physical reunions impossible.

School Legacy addresses this gap by combining generative AI with social Extended Reality (XR) to transform memory from a static archive to a living, shared experience. The system enables users to collaboratively reconstruct their classroom environment, generate personalized memory artifacts through AI, and engage in synchronized social rituals within a shared virtual space, turning individual nostalgia into a collective embodied reunion.

2 System Design

School Legacy comprises three core components that work together to deliver a comprehensive mixed reality (MR) and virtual reality (VR) experience for memory reconstruction.

2.1 User Experience Flow

The user journey progresses from physical to virtual collaboration: **Scan & Place** → Users scan QR codes to place virtual classroom

objects in their environment via MR; **Transform Memory** → Personal school artifacts are photographed and converted to 3D time capsules through AI; **Customize Avatars** → Users personalize avatars for social interaction; **Engage in Rituals** → Users enter the shared VR classroom for nostalgic activities; **Capture & Share** → Gesture recognition enables group photos as stylized Polaroid snapshots.

2.2 Three Core Components

Spatial Memory Construction. The experience begins with a printed memory book containing embedded QR codes for classroom elements such as the desk, locker, and student. Using MR passthrough on Meta Quest 3, users scan codes to spatially anchor 3D classroom objects in their physical environment. This tangible interaction grounds the digital experience in familiar, ritual-like behavior, transforming their current space into a foundation for memory reconstruction. The approach builds on established frameworks for tangible interaction design that emphasize physical space as a support for narrative engagement [Hornecker and Buur 2006].

AI-Powered Memory Transformation. The system's generative AI core transforms personal 2D photographs of school artifacts into textured 3D "time capsules" using the Tripo3D API [Yuan et al. 2023] [Yao et al. 2025]. This process democratizes the creation of 3D content, enabling it without technical expertise while maintaining emotional authenticity. Each generated object carries personal significance and functions as both a memorial artifact and an interactive key to enter the shared VR environment.

The AI pipeline processes uploaded images by automatically removing backgrounds and isolating objects, then generates 3D meshes with texture mapping. The generated assets are optimized for real-time VR rendering and support intuitive grasping and interaction [Xiong et al. 2025]. Unlike systems that treat AI output as final products, School Legacy positions AI as a collaborative layer where content is generated individually but gains meaning through shared social use.

Collective Memory Engagement. Once users enter the shared VR classroom through their personalized time capsules, they participate in synchronized social rituals designed to recreate authentic school experiences. The central projector serves as a ceremonial hub where users insert their "memory reel," a symbolic cartridge containing their graduation cap, time capsule, and yearbook contributions. This design draws on research into emotional heritage and digital nostalgia, reframing memory as an active, social practice rather than passive consumption [Turkle 2011].

The multiplayer environment supports co-presence through synchronized avatars, voice communication, and gesture tracking, aligning with established frameworks for social presence in virtual environments [Biocca et al. 2003]. Core interaction experiences include *Time Capsule Sharing*, where users present their AI-generated objects and personal stories, *Collaborative Graduation Cap Decoration* for real-time co-creation, *Digital Yearbook Signing* with voice-to-text message recording, *Nostalgic Music Sharing* through a virtual iPod interface, *Collective Drawing* on shared chalkboards, and gesture-activated *Group Photos Captured* as stylized Polaroid snapshots. (As shown in Figure 1).

3 Technical Implementation

The system architecture combines multiple advanced technologies into a unified pipeline. The foundation consists of Unity 6 with the Meta XR SDK for Quest 3 compatibility. MR functionality leverages Meta Quest passthrough with the MR Utility Kit for spatial anchoring of virtual objects. AI integration utilizes the Tripo3D API for real-time 2D-to-3D conversion with automated texture generation. Multiplayer synchronization employs Photon Fusion networking for real-time co-presence and state management across multiple users. Voice processing capabilities use the Meta Voice SDK for natural language input and yearbook message capture. Gesture recognition systems implement custom scripts for timed photo capture and social interaction mechanics. Key challenges included real-time lighting for accurate material rendering, synchronizing user states across transitions, and ensuring a smooth user experience in both solo and collaborative modes.

4 User Study

We conducted an informal user study with 6 group members to evaluate usability, emotional engagement, and collaborative dynamics. Participants reported strong emotional connections to AI-generated time capsules and a heightened sense of social presence, with 5 of 6 describing the experience as "emotionally meaningful." However, we identified challenges with synchronization timing and onboarding complexity. Building on these insights, we will conduct a formal study with 15 participants in groups of 3–5, including pre-session surveys, 45-minute guided experiences, post-session interviews, and two-week follow-up surveys to analyze long-term impact on memory reconstruction and social connection.

5 Conclusion and Future Work

School Legacy redefines nostalgia as a living, shared practice. By blending tangible artifacts, AI-generated 3D memories, and social XR, it turns recollection into co-creation. While rooted in the classroom, its modular design scales to homes, neighborhoods, and lost worlds, reconstructed not from perfect data, but from emotional fragments. Imagine displaced communities rebuilding villages in VR, with AI reconstructing architecture and simulating seasonal rhythms. Or schools archiving each graduating class not as a yearbook, but as an evolving memory world, updated annually as alumni contribute new stories and AI-generated retrospectives. School Legacy shows how immersive technologies can help us not just remember the past, but live in it, together.

References

- Frank Biocca, Chad Harms, and Judee K. Burgoon. 2003. Toward a more robust theory and measure of social presence: Review and suggested criteria. *Presence: Teleoperators & Virtual Environments* 12, 5 (2003), 456–480. MIT Press, Cambridge, MA, USA. <https://doi.org/10.1162/09636620322753522> (2003).
- Eva Hornecker and Jacob Buur. 2006. Getting a grip on tangible interaction: A framework on physical space and social interaction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '06)*. Association for Computing Machinery, New York, NY, USA, 437–446. doi:10.1145/1124772.1124840
- Sherry Turkle. 2011. *Alone Together: Why We Expect More from Technology and Less from Each Other*. Basic Books, New York, NY, USA (2011).
- Haowei Xiong, Kexin Nie, Jiachen Zeng, Shujing Shen, and Mengyao Guo. 2025. SugART: Mixed Reality Sugar Painting for Intangible Cultural Heritage Learning at Home. In *Proceedings of the Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters '25)*. Association for Computing Machinery, New York, NY, USA, Article 26, 3 pages.

doi:10.1145/3721250.3742997
Zhihao Yao, Xiwen Yao, Qirui Sun, Haowei Xiong, Kin Wang Lau, and Haipeng Mi.
2025. Gesture2Prop: Conditional VR Prop Generation from Hand Gestures and
Speech. In *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface*

Software and Technology (UIST Adjunct '25). Association for Computing Machinery,
New York, NY, USA, Article 38, 4 pages. doi:10.1145/3746058.3759017
Lu Yuan, Qiang Chen, Yuxuan Xiong, Hao Pan, and Chengquan Zhang. 2023. Tripo:
Fast 3D reconstruction from a single image. *arXiv preprint arXiv:2311.17464* (2023).